

6. Run on one patient (single test)

```
[40] result = run_triage_agent(PATIENTS[0])
```

```
[TOOL CALL] calculate_news2_score {'hr': 130, 'systolic_bp': 82, 'resp_rate': 28, 'spo2': 89, 'gcs': 13, 'temp_c': 38.9}
```

```
[TOOL RESULT] {'score': 15, 'tier': 'CRITICAL'}
```

```
[TOOL CALL] find_and_reserve_bed {'tier': 'CRITICAL', 'lat': 19.99, 'lon': 73.785}
```

```
[TOOL RESULT] {'status': 'RESERVED', 'hospital_id': 'H1', 'hospital_name': 'City General Hospital', 'bed_type': 'ICU', 'distance_km': 0.97, 'remaining_beds': 2}
```

```
[AGENT SUMMARY]
```

```
Patient scored 15/20 -> priority tier CRITICAL. Reserved a ICU bed at City General Hospital (0.97 km away). 2 ICU beds remaining there.
```

7. Full demo — all patients

```
41] with open("hospitals.json") as f:
3s   HOSPITALS = json.load(f)

for p in PATIENTS:
    print()
    print("--- Incoming:", p["name"], "(" + p["id"] + ") ---")
    print("Description:", p["description"])
    run_triage_agent(p)
    time.sleep(1)

print()
print("=== Demo complete ===")
```

```
...
--- Incoming: Patient A (P1) ---
Description: Motorcycle accident, visible chest trauma, labored breathing
[TOOL CALL] calculate_news2_score {'hr': 130, 'systolic_bp': 82, 'resp_rate': 28, 'spo2': 89, 'gcs': 13, 'temp_c': 38.9}
[TOOL RESULT] {'score': 15, 'tier': 'CRITICAL'}
[TOOL CALL] find_and_reserve_bed {'tier': 'CRITICAL', 'lat': 19.99, 'lon': 73.785}
[TOOL RESULT] {'status': 'RESERVED', 'hospital_id': 'H1', 'hospital_name': 'City General Hospital', 'bed_type': 'ICU', 'distance_km': 0.97, 'remaining_beds': 2}
[AGENT SUMMARY]
Patient scored 15/20 -> priority tier CRITICAL. Reserved a ICU bed at City General Hospital (0.97 km away). 2 ICU beds remaining there.

--- Incoming: Patient B (P2) ---
Description: Fall from ladder, suspected wrist fracture, alert and stable
[TOOL CALL] calculate_news2_score {'hr': 95, 'systolic_bp': 118, 'resp_rate': 18, 'spo2': 97, 'gcs': 15, 'temp_c': 37.1}
[TOOL RESULT] {'score': 1, 'tier': 'STANDARD'}
[TOOL CALL] find_and_reserve_bed {'tier': 'STANDARD', 'lat': 19.97, 'lon': 73.77}
[TOOL RESULT] {'status': 'RESERVED', 'hospital_id': 'H2', 'hospital_name': "St. Mary's Medical Center", 'bed_type': 'General', 'distance_km': 1.11, 'remaining_beds': 14}
[AGENT SUMMARY]
Patient scored 1/20 -> priority tier STANDARD. Reserved a General bed at St. Mary's Medical Center (1.11 km away). 14 General beds remaining there.
```

```

... --- Incoming: Patient C (P3) ---
Description: Cardiac arrest resuscitated en route, unresponsive, unstable
[TOOL CALL] calculate_news2_score {'hr': 145, 'systolic_bp': 70, 'resp_rate': 32, 'spo2': 84, 'gcs': 9, 'temp_c': 36.5}
[TOOL RESULT] {'score': 15, 'tier': 'CRITICAL'}
[TOOL CALL] find_and_reserve_bed {'tier': 'CRITICAL', 'lat': 20.005, 'lon': 73.75}
[TOOL RESULT] {'status': 'RESERVED', 'hospital_id': 'H3', 'hospital_name': 'Riverside Trauma Center', 'bed_type': 'ICU', 'distance_km': 0.7, 'remaining_beds': 4}
[AGENT SUMMARY]
Patient scored 15/20 -> priority tier CRITICAL. Reserved a ICU bed at Riverside Trauma Center (0.7 km away). 4 ICU beds remaining there.

--- Incoming: Patient D (P4) ---
Description: Minor lacerations from glass, ambulatory, anxious but stable
[TOOL CALL] calculate_news2_score {'hr': 88, 'systolic_bp': 122, 'resp_rate': 16, 'spo2': 98, 'gcs': 15, 'temp_c': 36.9}
[TOOL RESULT] {'score': 0, 'tier': 'STANDARD'}
[TOOL CALL] find_and_reserve_bed {'tier': 'STANDARD', 'lat': 19.94, 'lon': 73.805}
[TOOL RESULT] {'status': 'RESERVED', 'hospital_id': 'H4', 'hospital_name': 'Nashik District Hospital', 'bed_type': 'General', 'distance_km': 0.94, 'remaining_beds': 11}
[AGENT SUMMARY]
Patient scored 0/20 -> priority tier STANDARD. Reserved a General bed at Nashik District Hospital (0.94 km away). 11 General beds remaining there.

=== Demo complete ===

```

8. Bonus: predictive deterioration check

```

[42]
✓ 0s
simulated_stream = [
    {"spo2": 95, "hr": 90, "t": 0},
    {"spo2": 93, "hr": 100, "t": 60},
    {"spo2": 89, "hr": 118, "t": 120},
]
print(json.dumps(check_deterioration_trend(simulated_stream), indent=2))

```

```

... {
  "alert": true,
  "reason": "SpO2 dropped 6 pts / HR rose 28 bpm"
}

```